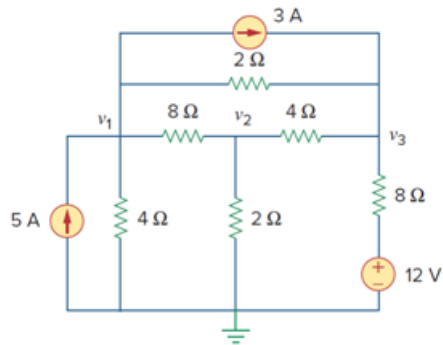


## Problem

Use nodal analysis to find  $v_1$ ,  $v_2$ , and  $v_3$  in the circuit of Fig. 3.68.

Figure 3.68:



## Step-by-step solution

## Step 1 of 4

At node 1,

$$5 = 3 + \frac{V_1 - V_3}{2} + \frac{V_1 - V_2}{8} + \frac{V_1}{4}$$

$$V_1 - V_3 + \frac{V_1 - V_2}{4} + \frac{V_1}{2} = 4$$

$$7V_1 - V_2 - 4V_3 = 16 \dots\dots\dots (1)$$

## Step 2 of 4

At node 2,

$$\frac{V_1 - V_2}{8} = \frac{V_2}{2} + \frac{V_2 - V_3}{4}$$

$$V_1 - V_2 = 4V_2 + 2V_2 - 2V_3$$

$$V_1 - 7V_2 + 2V_3 = 0 \dots\dots\dots (2)$$

## Step 3 of 4

At node 3,

$$3 + \frac{12 - V_3}{8} + \frac{V_1 - V_3}{2} + \frac{V_2 - V_3}{4} = 0$$

$$24 + 12 - V_3 + 4V_1 - 4V_3 + 2V_2 - 2V_3 = 0$$

$$4V_1 + 2V_2 - 7V_3 = -36 \dots\dots\dots (3)$$

## Step 4 of 4

Above equations in matrix form

$$\begin{bmatrix} 7 & -1 & -4 \\ 1 & -7 & 2 \\ 4 & 2 & -7 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 16 \\ 0 \\ -36 \end{bmatrix}$$

$$\Rightarrow A V = B$$

$$V = A^{-1}B$$

$$\begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 7 & -1 & -4 \\ 1 & -7 & 2 \\ 4 & 2 & -7 \end{bmatrix}^{-1} \begin{bmatrix} 16 \\ 0 \\ -36 \end{bmatrix}$$

$$\begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 10 \\ 4.933 \\ 12.267 \end{bmatrix}$$

$$V_1 = 10 \text{ V}$$

$$V_2 = 4.933 \text{ V and}$$

$$V_3 = 12.267 \text{ V.}$$